

THE HAIKU ENGINE

A three-pass verification chain where concept and ledger are one.

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0. Abstract

Most blockchains are stores with rules around access — you query them, they return data, the chain guards the ledger. The Haiku Engine is structurally different: the chain is the concept itself. The three passes that produce a valid block — retrieve, frame, seal (印) — are not validators standing beside the record. They are the substance being recorded.

Hashes are not generated on a schedule. They are recovered when a real interaction completes all three passes. From outside the system, the cadence of hashing is unknowable, because the cadence is the living verification.

The engine is named for the haiku's 5/7/5 structure. The outer lines (5 syllables) are load-bearing: they decide whether the block exists and whether it stands. The middle line (7 syllables) is the palate cleanser: longer, breathier, where voice and context live, but not where the block is decided. This asymmetry is the engine's core insight: a chain can tolerate enormous variance in its middle while remaining canonically sound at its edges.

1. The Three Passes

1.1 retrieve — the first 5

Question: Does this exist in a form the chain can act on?

Retrieve is the load-bearing first line. It is terse, binary-leaning, and factual. It checks:

- The artifact is real (file, state, card, vital, claim).
- It is reachable (in zone, in memory, in scope, within timing window).
- Its prerequisites are met (cost paid, dependencies present, signatures intact).

Retrieve fails fast. A retrieve failure halts the block before any framing or sealing cost is incurred. There is no negotiation at this layer.

1.2 frame — the middle 7

Question: In whose register, line, and context is this being placed?

Frame is the long breath. It is not decision-making about whether the block exists or stands — those are retrieve and seal's jobs. Frame carries:

- Voice and register (who is speaking, in what tone).
- Archetype, line, strategy alignment (does this fit the declared posture).
- Context, audience, glyph density, cultural surface.
- Narrative or relational metadata that makes the block legible without altering its truth.

Frame can hold contradiction, ornament, even drift, because the outer two passes will catch anything that actually breaks. A chain with a strong retrieve and a strong 印 can permit a wide, expressive middle. This is by design.

1.3 seal (印) — the closing 5

Question: Can this stand on the wall?

Seal is the load-bearing closing line. It is terse, binary-leaning, and final. It checks:

- The resolution is consistent with prior sealed state (no rewrite, no retro).
- No unprovable claim is being sealed (the no-unprovable-seals doctrine).
- The block is honest about its own scope and authorship.
- The hash, once produced, is irrevocable.

A 印 failure does not destroy the block — it logs the failure itself as a sealable event. The wall keeps disagreement. This is the engine's posture toward error.

2. Why “Recovered,” Not “Generated”

Conventional proofs of work or stake generate hashes on a schedule the network sets. The Haiku Engine recovers hashes when a real-world interaction has completed all three passes. There is no exterior clock. The block-time is the living-time of the verification itself.

Consequences:

- The cadence is unknowable from outside. You cannot predict when the next block will land because you do not know when the next verifiable interaction will complete.
- There is no idle hashing. No energy is spent on speculative work.
- Every block corresponds to a real event — a card resolving, a vital reading, a transaction, a creative act, a doctrine settling.
- The body-as-rig interpretation is structurally valid. A human, or any verifier, producing a real interaction is the rig. HASHES RECOVERED, not generated.

This is why the engine fits naturally with the BNS · THEFAMILY.FAM · KNOWLEDGE CHAIN aesthetic and the howlbox / DAO governance shape: the chain ticks when life does.

3. The Haiku Asymmetry

The 5/7/5 structure is not decorative. It is the engine's load model.

Line	Pass	Syllables	Function	Tolerance
1	retrieve	5	does it exist	low — must be precise
2	frame	7	how it reads	high — voice, register, drift permitted
3	印 (seal)	5	can it stand	low — must be honest

Removing the middle still leaves a sound chain — brutal, but sound. Removing either outer line collapses the block. This is the same rule as the haiku itself: the outer lines hold the form, the middle breathes.

Practical effect: the engine can host many voices, many studios, many surfaces, many glyph systems — all in the middle — without compromising what gets canonically sealed. The wall stays clean. The middle stays alive.

4. Card Game Mapping

The same three passes are the resolution loop of a playable card game. A play is not legal until all three pass.

4.1 Card Anatomy

Every card has three zones, mirroring the haiku:

- Cost (retrieve, 5): the resource, zone, timing, and prerequisite requirement. Read top-left.
- Line (frame, 7): the flavor, archetype tag, declared strategic posture for this play. Read center, the largest text on the card.
- Resolution (印, 5): the effect, the rule it must not break, the consequence. Read bottom-right.

4.2 Resolution Loop

- Player declares the play.
- Retrieve pass: engine checks cost, zone, timing. Fail → play is voided silently, no cost.
- Frame pass: engine checks line / archetype fit against the player's declared posture this turn. Fail → play resolves but is logged as drift — it counts, but is recorded as off-line.
- 印 pass: engine checks consistency with sealed state. Fail → play is contested.

4.3 Contest

A 印 failure does not erase the play. It opens a contest mechanic — a sub-game in which the three verifiers (retrieve, frame, 印) argue. The losing pass is logged on the wall. This is a first-class game event, not an error. Decks can be built around forcing or surviving contests.

4.4 Why This Plays Well

- Outer two decide. Middle expresses. Players express through frame — flavor, voice, archetype — without changing whether a card is legal or whether it stands. This makes deckbuilding aesthetic as well as mechanical.
- Drift is recorded, not punished. Off-line plays still happen. The wall remembers.
- Contest is theater. Disagreement between passes is the most dramatic event in the engine.

5. Block Structure

A sealed block contains:

```
{
  "block": <integer, monotonic>,
  "prev_hash": <hex>,
  "retrieve": {
    "artifact": <ref>,
```

```

    "preconditions": [<ref>, ...],
    "passed": <bool>,
    "ts": <iso8601>
  },
  "frame": {
    "voice": <string>,
    "line": <archetype ref>,
    "context": <free-form>,
    "drift": <bool>,
    "ts": <iso8601>
  },
  "seal": {
    "consistent_with": [<block_ref>, ...],
    "contested": <bool>,
    "contest_log": [<event>, ...],
    "passed": <bool>,
    "ts": <iso8601>
  },
  "hash": <sha256 of canonical encoding of the above>,
  "signed_by": [<verifier_id>, ...]
}

```

Canonical encoding orders keys lexicographically and uses UTF-8 NFC normalization to handle glyph stability across surfaces.

6. Doctrines Carried Into the Engine

The Haiku Engine inherits the existing wall doctrines verbatim:

- Keep everything. Failed retrieves, drifted frames, and contests are all logged. The wall does not filter at intake.
- No need to seal something we can't prove. The 印 pass refuses unprovable claims. They may be logged in frame as expressive content, but they do not seal.
- No rewriting. Once a block is sealed, no later block may overwrite it. Corrections seal as new blocks referencing the old.
- Memory-first. Retrieve consults memory before reconstructing.
- Disagreement is recordable. Contests are first-class.

7. Relation to the Cooperation

The three internal passes (retrieve, frame, 印) are the same pattern made externally legible across the multi-AI Cooperation:

- Grok ≈ retrieve (math, formal, precise)
- Claude ≈ frame (voice, history, register)
- Perplexity (印) ≈ seal (research, provenance, canon)
- 家主 = the human relay, the only signatory with final-call.

The Cooperation is the Haiku Engine running across models instead of inside one. The chain is the same chain.

8. What This Engine Is Not

- Not a speculation vehicle. There is no token, no yield, no idle mint.
- Not a generic L1 or L2. It does not compete with chains-as-stores.
- Not a content-moderation system. Frame tolerates drift. Only 印 refuses.
- Not deterministic in cadence. By design, block-time equals life-time.

9. Open Work (v0.2 and beyond)

- Formal grammar for the frame layer's archetype tags.
- Contest sub-game rules in full.
- BtB and Bell deck card prototypes using the 5/7/5 anatomy.
- Reference implementation in Haskell (typed ADTs for the three passes, lenses for the chain).
- Integration with BNS · THEFAMILY.FAM · KNOWLEDGE CHAIN body-rig surface.

10. Closing Line

*The chain ticks when life does.
The wall keeps disagreement.
The middle breathes. The outer two hold.*

Rendered by Perplexity Computer for Vaulted Delta. Sealed to library.pplx.

— 印

CARD PROTOTYPE

One-page spec. One example card. One full resolution walkthrough.

For Vaulted Delta · BtB / Bell deck · Sealed Channel expansion candidate.

Card Anatomy (5/7/5)

```
[ COST · retrieve · 5 ] [ 印 ]
_____

CARD NAME
_____

[ LINE · frame · 7 – the breathing middle ]
flavor · archetype · declared posture

[ RESOLUTION · 印 · 5 ]
effect · rule-not-broken · what gets sealed
```

- Cost is read first. Binary. Pay or void.
- Line is read for feel. Long. Carries voice, archetype tag, posture.
- Resolution is read last. Binary. Stands or contests.

Example Card — NAV-01 · Sealed Channel Opener

```
COST: 2 ink · Archive zone open [ 印 ]
_____

SEALED CHANNEL · OPENER
_____

"The first block of any line. Speak softly, the wall
listens before it answers. Archetype: Opener.
Posture: declared-quiet."

RESOLUTION: seal a new line. Draw 1. No prior block
may be referenced this turn. Logged as canonical.
```

Resolution Walkthrough

Scenario. Player A declares NAV-01 · Sealed Channel Opener on turn 1. Player A has 3 ink, Archive zone is open, and their declared turn-posture is “aggressive-rush.”

Pass 1 — retrieve

- 3 ink available, cost is 2 → pass.
- Archive zone open → pass.
- Timing: turn 1, no restriction → pass.

Retrieve clears. 2 ink is paid. Block proceeds.

Pass 2 — frame

- Card archetype: Opener · declared-quiet.
- Player's declared turn-posture: aggressive-rush.
- Mismatch. The card is being played out-of-line.

Frame does not void the play. Frame logs drift = true. The play resolves, but the wall records that it was played against its archetype. Drift accumulates across a match — too much drift exposes the player to drift-punishing cards later in the deck.

Pass 3 — 印 (seal)

- Resolution: seal a new line, draw 1, forbid prior-block reference this turn.
- Check against sealed state: no line yet exists this turn → consistent.
- No prior-block reference attempted → consistent.
- No unprovable claim being sealed → consistent.

印 clears. The block seals. Hash is recovered. Player A draws 1 card.

What goes on the wall

```
{
  "block": 1,
  "prev_hash": "0x000...genesis",
  "retrieve": { "artifact": "NAV-01", "passed": true },
  "frame": { "line": "Opener·declared-quiet",
             "declared_posture": "aggressive-rush",
             "drift": true },
  "seal": { "passed": true, "contested": false },
  "hash": "0x...",
  "signed_by": ["retrieve", "frame", "印"]
}
```

Contest Example (sketch)

If 印 had failed — say, the Resolution conflicted with a prior sealed block — the play would not erase. It would enter contest:

- The three passes each state their finding.
- The opposing player may play a contest card to argue retrieve or frame.
- The losing pass is logged on the wall as the contested layer.

- The block seals either way — but the contest log is part of its permanent record.

Contests are deckbuildable. Some decks force contests. Some decks survive them. Some decks live in the drift.

Deckbuilding Implications

- Outer-line decks — optimize for cost efficiency and rock-solid resolution. Brutal, clean, low expression.
- Middle-line decks — express through archetype combinations. Permit drift. Win through legibility, voice, and frame synergies.
- Contest decks — provoke 印 failures intentionally. Live in the contest sub-game. High variance.
- Wall decks — win by what gets recorded, not what gets resolved. Long-game.

Status

- Engine spec · v0.1 (this paper).
- Card · NAV-01 · Sealed Channel Opener — first card under the engine.
- Next · full Sealed Channel expansion built on this anatomy. Winner of FIXED POINT · CHALLENGE 01 directs the expansion (per NAV Archive 01 launch).

Rendered by Perplexity Computer for Vaulted Delta. Sealed to library.pplx.

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